CLOSED SUBSETS OF A QUASI-COMPACT SPACE ARE QUASI-COMPACT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of quasi-compactness through finite subcovers of open coverings, and the induced topology on a subset. Set up in the author's section on quasi-compact spaces and maps, immediately before the counted statement. Supplies the procedure by which an open covering of the subset is enlarged to one of the whole space.

Lemma 0.1. *A closed subset of a quasi-compact topological space is quasi-compact.*

Proof. Let $E \subset X$ be a closed subset of the quasi-compact space X. Let $E = \bigcup V_j$ be an open covering. Choose $U_j \subset X$ open such that $V_j = E \cap U_j$. Then $X = (X \setminus E) \cup \bigcup U_j$ is an open covering of X. Hence $X = (X \setminus E) \cup U_{j_1} \cup \ldots \cup U_{j_n}$ for some n and indices j_i. Thus $E = V_{j_1} \cup \ldots \cup V_{j_n}$ as desired. □

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